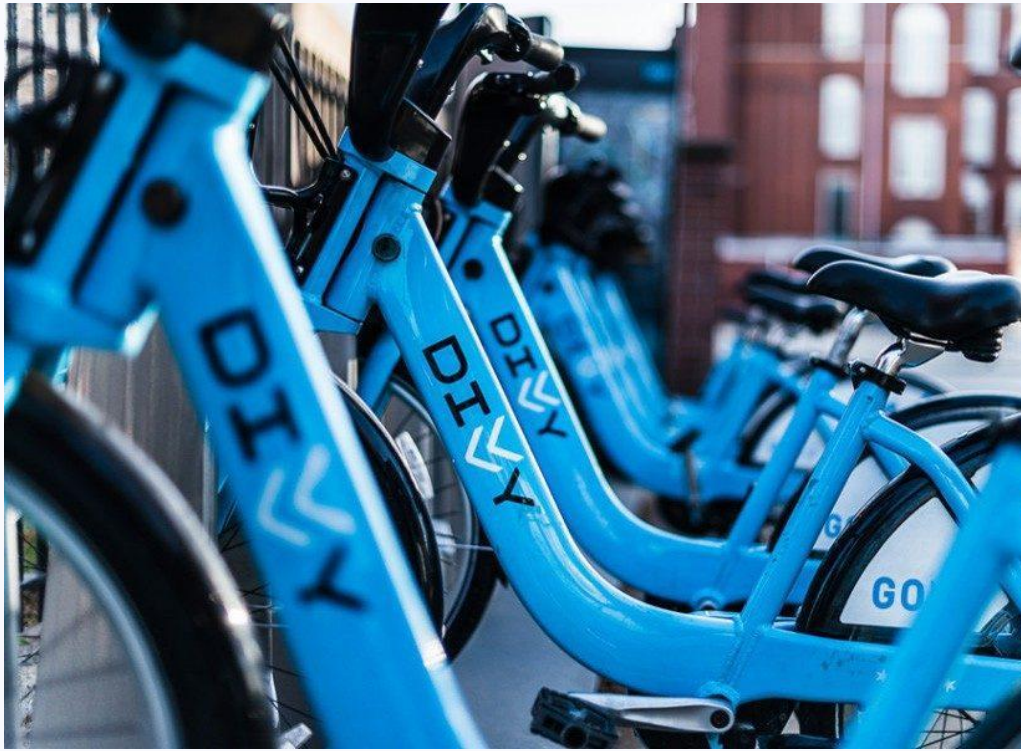
DIVVY CHICAGO BIKE SHARE ANALYSIS



Akhmad Fauzi Nurulhamzah



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BACKGROUND

Business Introductions

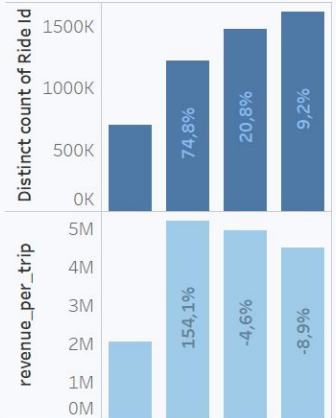
Divvy is Chicagoland's **bike share system across Chicago** and Evanston. Divvy provides bikes that can be unlocked from one station and returned to any other station in the system. The **primary source of revenue for Divvy is the fees charged for bike rentals**. Users pay for the duration of bike usage, which can vary based on the membership plan they choose.

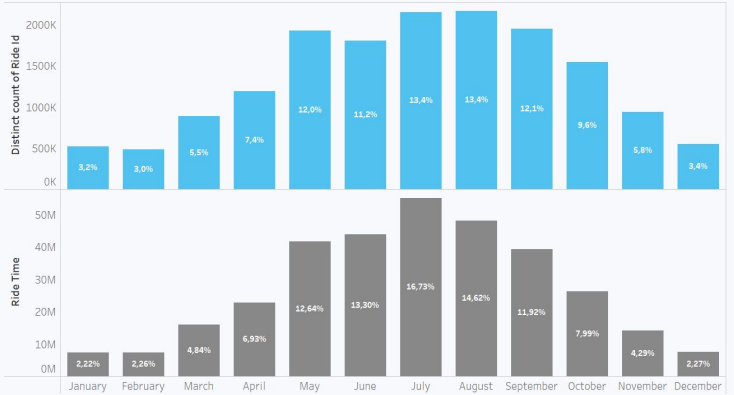
Objectives

The goal of this case study is to analyze the data and **create insights to develop strategy to increase annual revenue** for Divvy. As well as build dashboard from an important matrix to review the company performance

Problem Statement

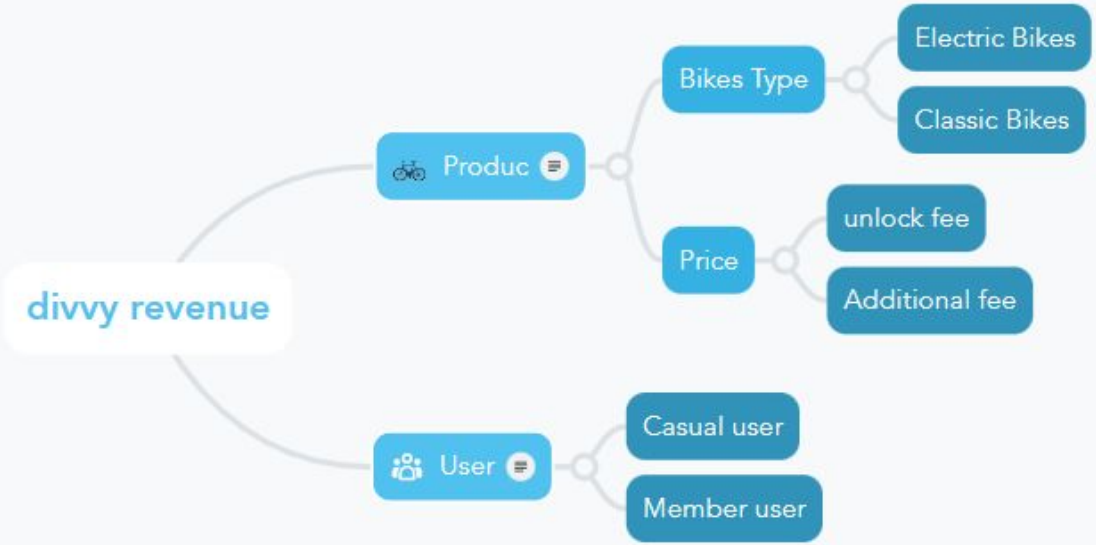
- Why the number of rides and ride duration were **low in the first quarter,** increased in the second and third quarters, yet **decreased once more in the fourth quarter.**
- Thought bike **ride number is increasing** by time, we find out that the sum of **daily revenue is decreasing** over time





BUSINESS QUESTIONS & RCA

- How often riders use the shared bikes within a given time frame?
- Which groups are the most frequent users of our shared bikes, and how are the characteristics of each group?
- Which type of bikes that are the most frequent users of our shared bikes, and how are the characteristics of each group?
- Are there specific areas or stations where our shared bikes are in high demand? How can we optimize bike distribution to meet this demand?
- What are the most popular trip durations, and how can we cater to the needs of riders based on their preferred trip lengths?
- Are there certain days or times when our shared bikes experience higher usage?



METHODOLOGY

Define Objectives

Determine the objective, scope, and root cause

1

Data Gathering

Conduct data gathering and information about bussiness

2

Data Preparation

Conduct data cleaning using Python

3

Data Analisys

Conduct Data Analisys Using Phyton and Tableu

4

Recomendations

Discuss suitable recommendations

5



EDA

Data Sizes

The analysis is based on data collected between **January 2020 and May 2023**. The dataset consists of two types of bicycles and two types of users.

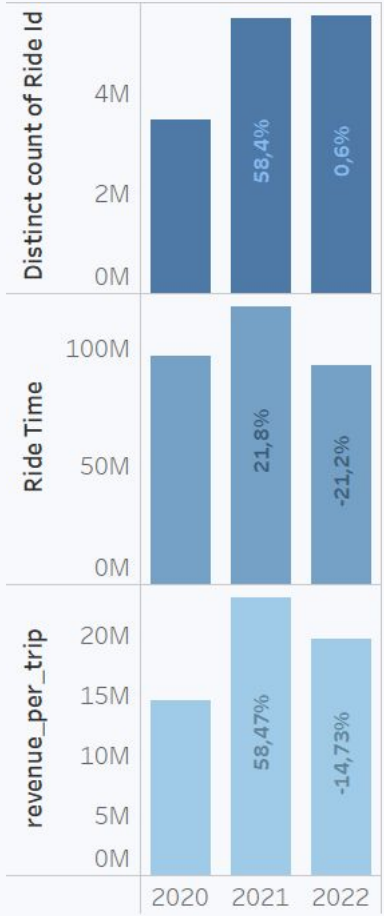
Ride Quantity	16,135,915
Ride Time	329.188.517

= **5,486,475.28 Jam**

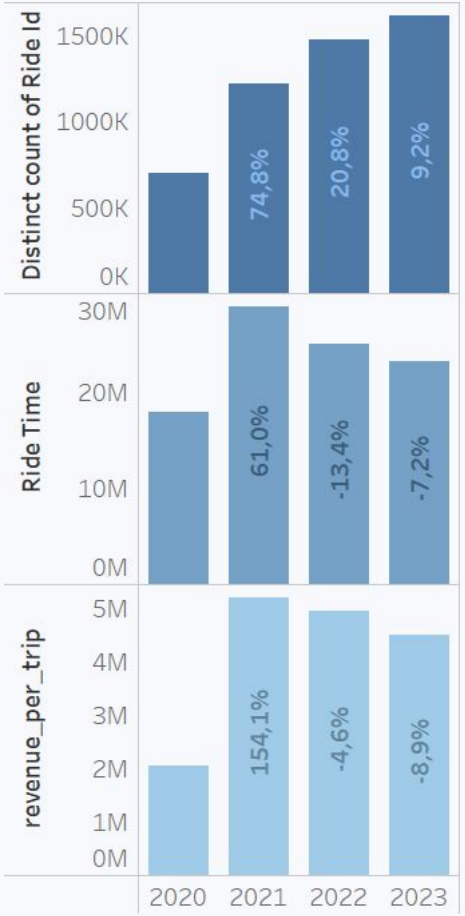
	Year			
	2020	2021	2022	2023
Ride Quantity	3,474,901	5,505,092	5,540,772	1,615,150
Ride Time	96.310.415	117.258.981	92.407.655	23.211.466

By observing the bar graph, it becomes evident that while the number of rides has been increasing annually, the duration of the rides is actually decreasing. It is important to note that **both the quantity of rides and their duration are crucial factors in determining the revenue**. As we see that the revenue per trip is also decreasing

Jan - Des



Jan - May



Bikes Type

Classic

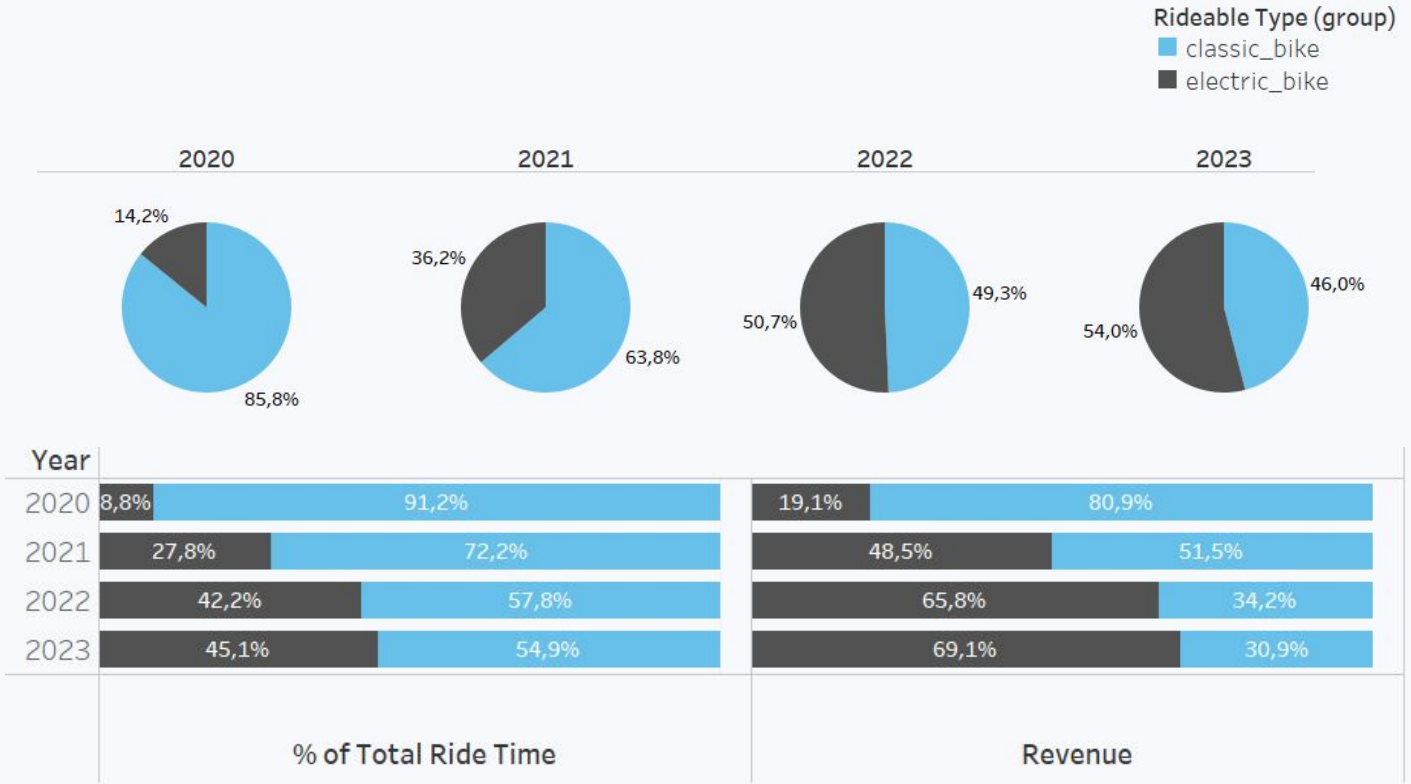


Member: 45 min free, then $ 0.17 /min
Casual: $ 1 unlock + $ 0.17/min

Electric



Member: $ 0.17/min
Casual: $ 1 unlock + $ 0.42/min



- **Electric bicycle usage has been steadily increasing** over the past few years, from 14.2% in 2020 to 54% in 2023.
- Electric bicycles are becoming more popular with users than classic bicycles.
- The higher rental fees for electric bicycles are not a problem for users.

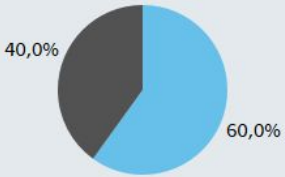
Members



Ride Quantity **Ride time** **Revenue**

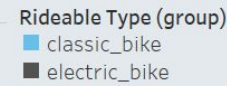
Casual User

	Year			
	2020	2021	2022	2023
Ride Quantity	1,341,775	2,491,420	2,268,975	509,501
Median Ride Time	22	16	13	11
Revenue	13,143,480	20,177,795	15,988,692	3,299,308

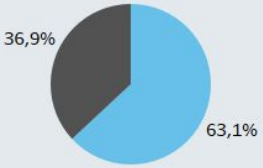


- In comparison to all rental actions, **59% are made by casual users.**
- Both the number of rentals and the duration of bike rentals show a **decline** from 2021.
- 60% of users prefer riding vintage bikes.

Member User

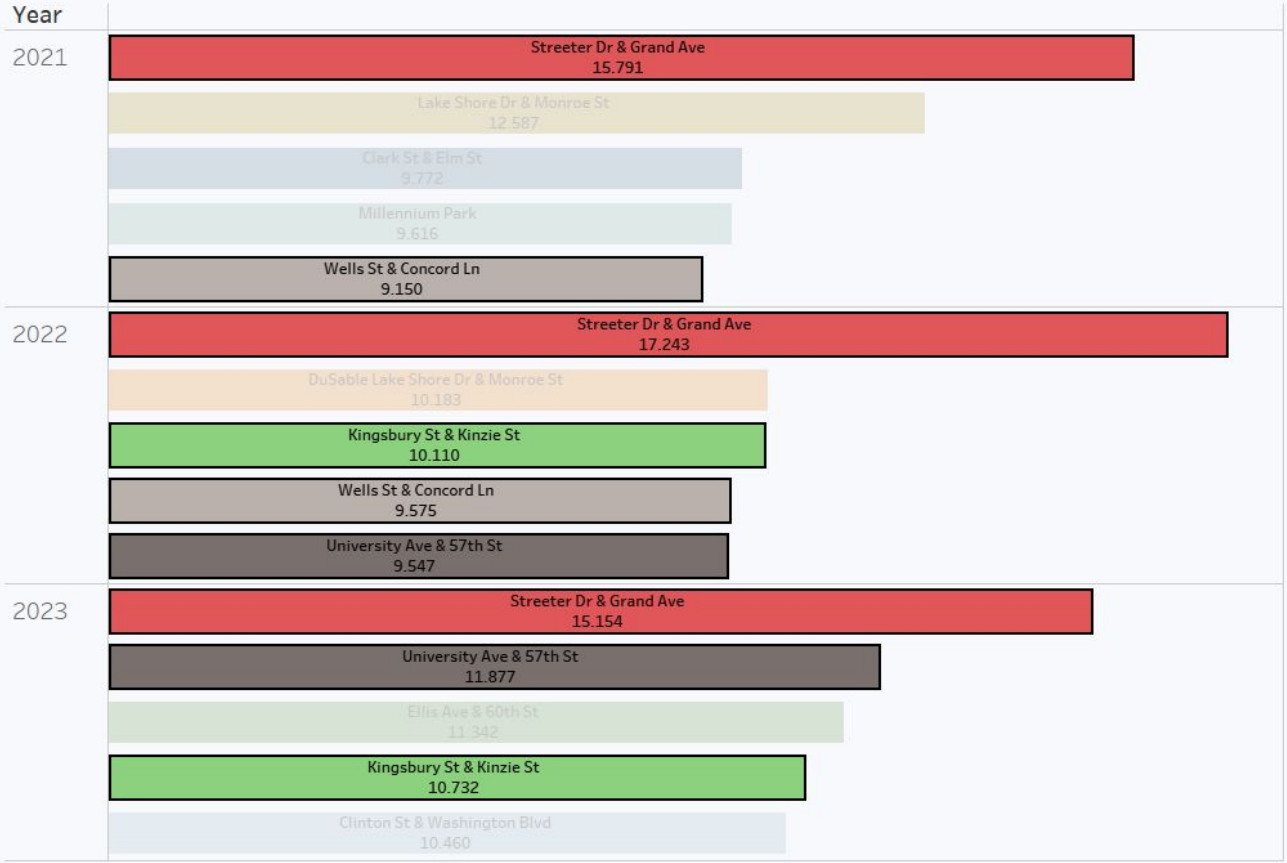


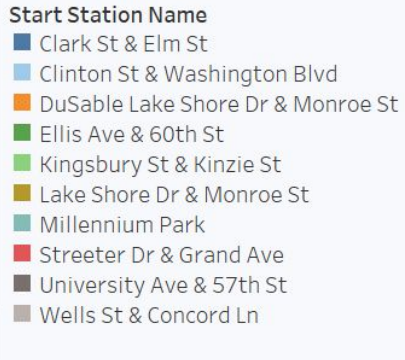
	Year			
	2020	2021	2022	2023
Ride Quantity	2,133,126	3,013,672	3,271,797	1,105,649
Median Ride Time	11	10	9	8
Revenue	1,384,387	2,844,875	3,642,523	1,205,026



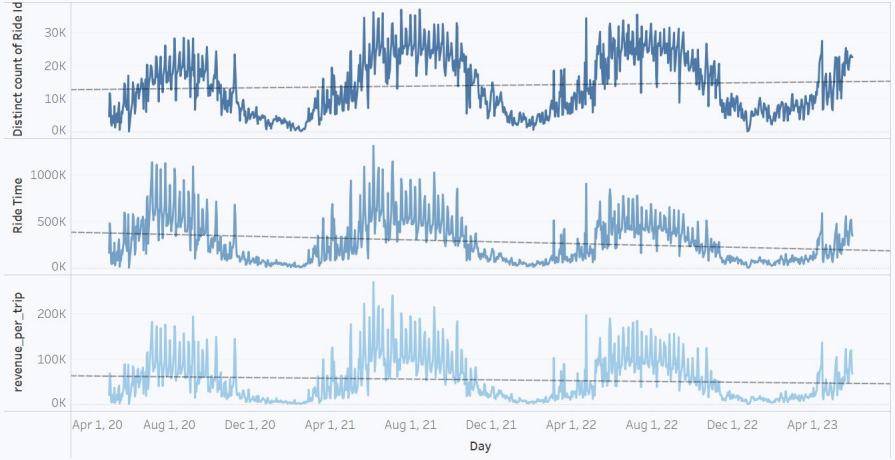
- Rentals made by member users made up 41% of all rental activity.
- Although the **quantity of rentals has grown** over time, the **timing of bike rentals suggests a reduction**.
- 63.1% of users indicated they like riding classic bikes.

Stations





- Streeter & Grand Ave, university ave, kingsbury st and wells st are popular stations that could be prioritized for additional investment.



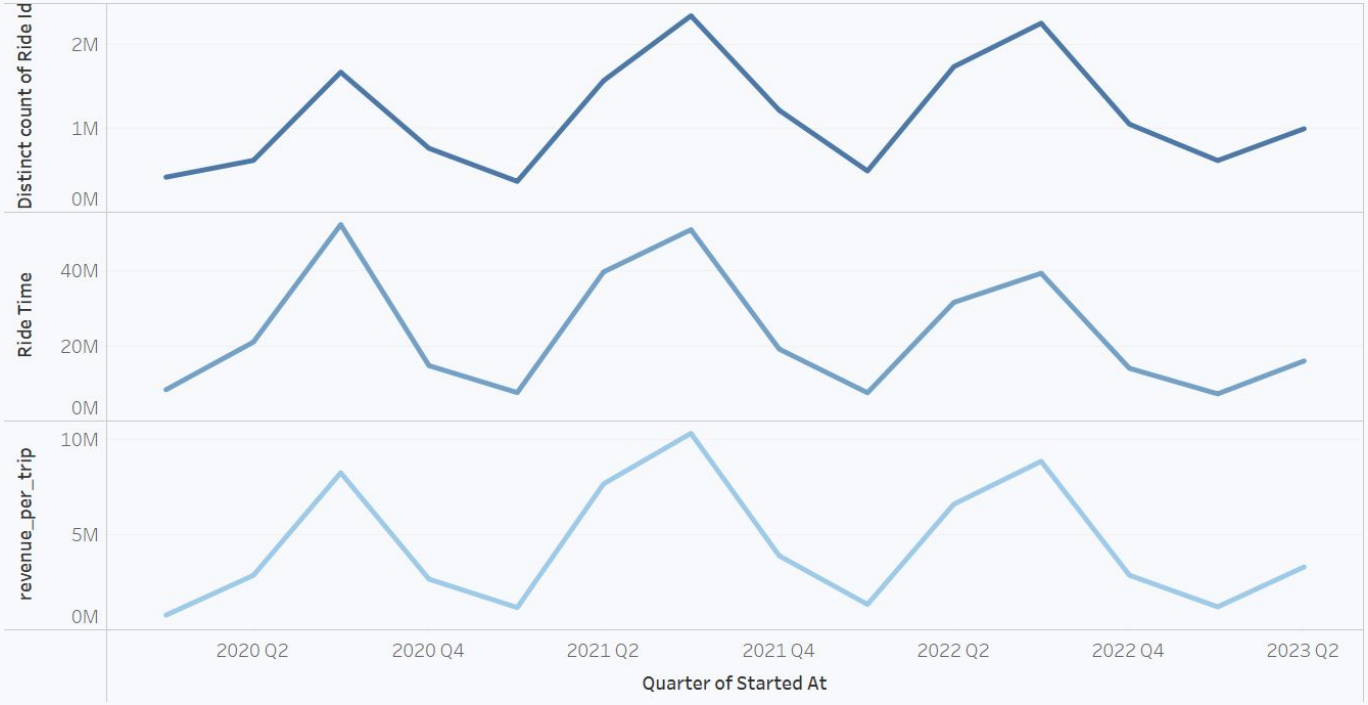
Timeseries Analysis



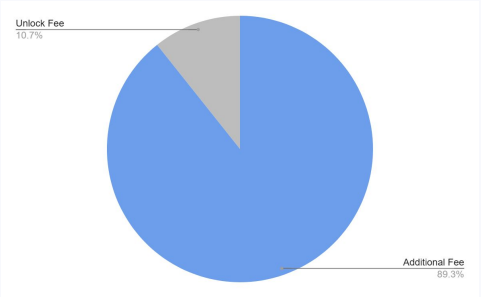




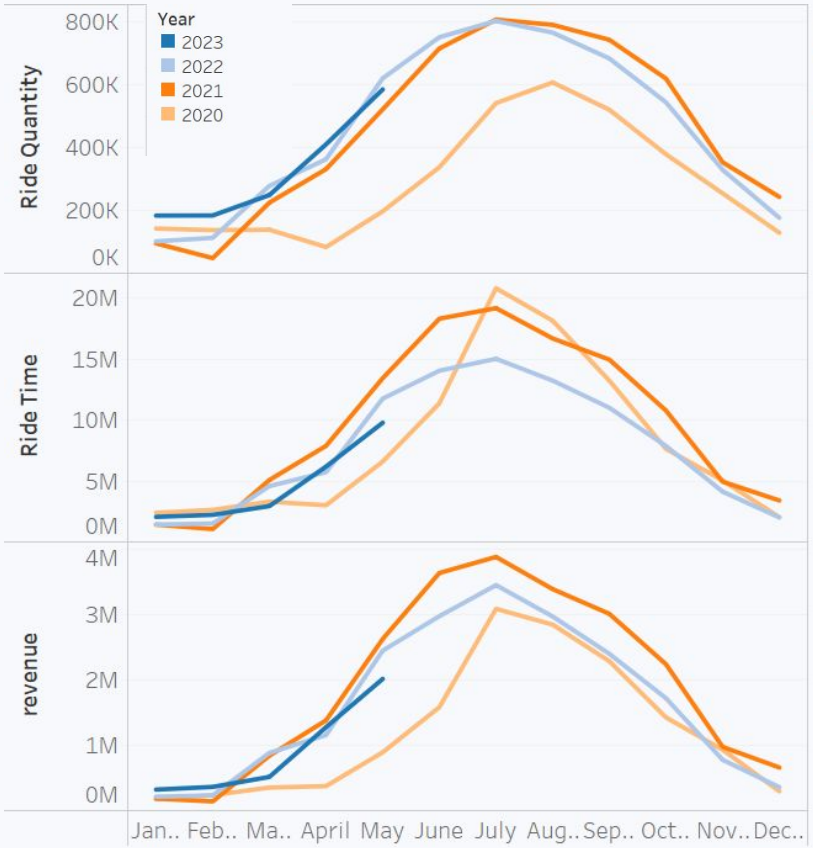
Divvy Bikes Trend

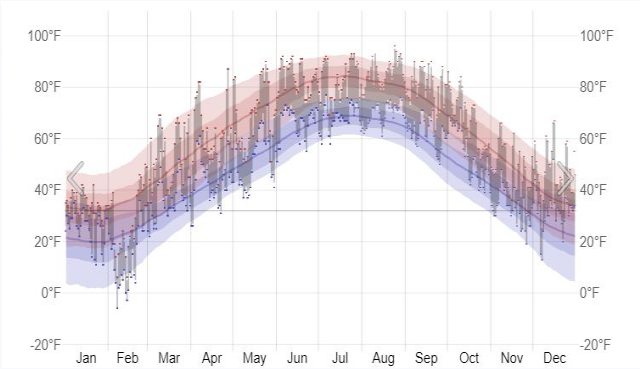


The timeseries data displays a **seasonal tendency,** as seen by the line chart, which happens in either ride quantity, ride time, and revenue.

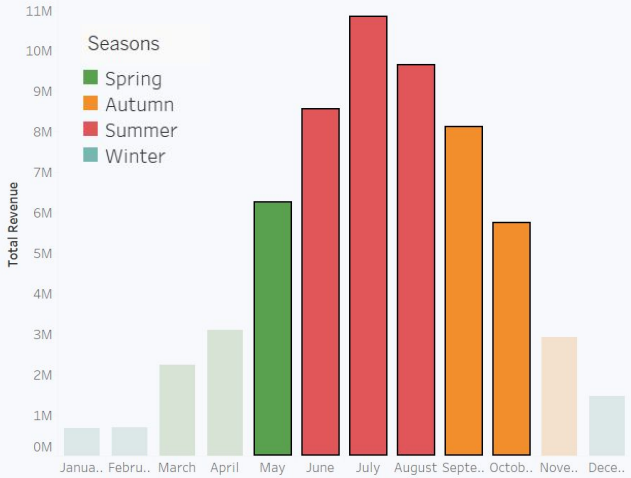


Per year comparations



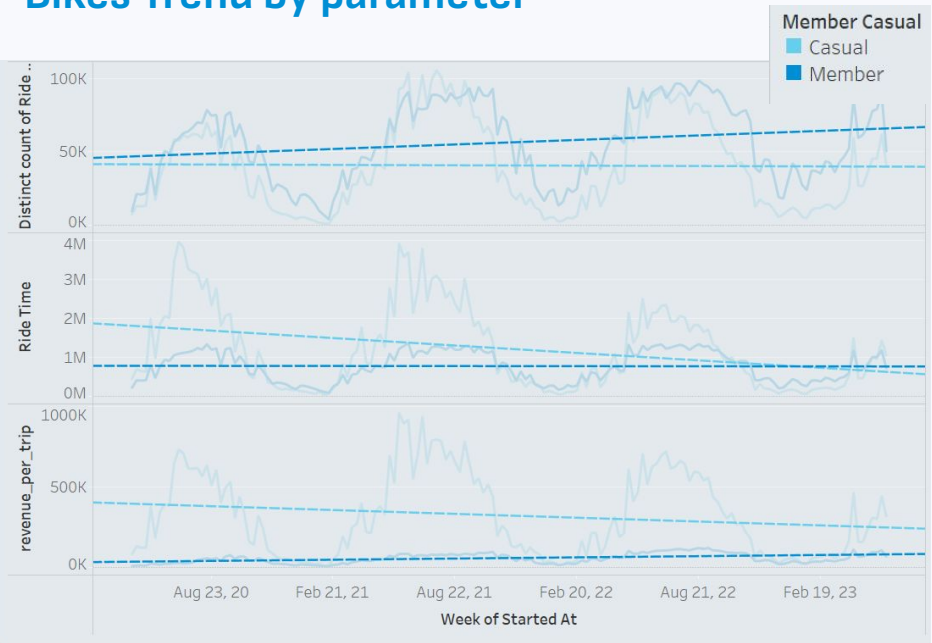


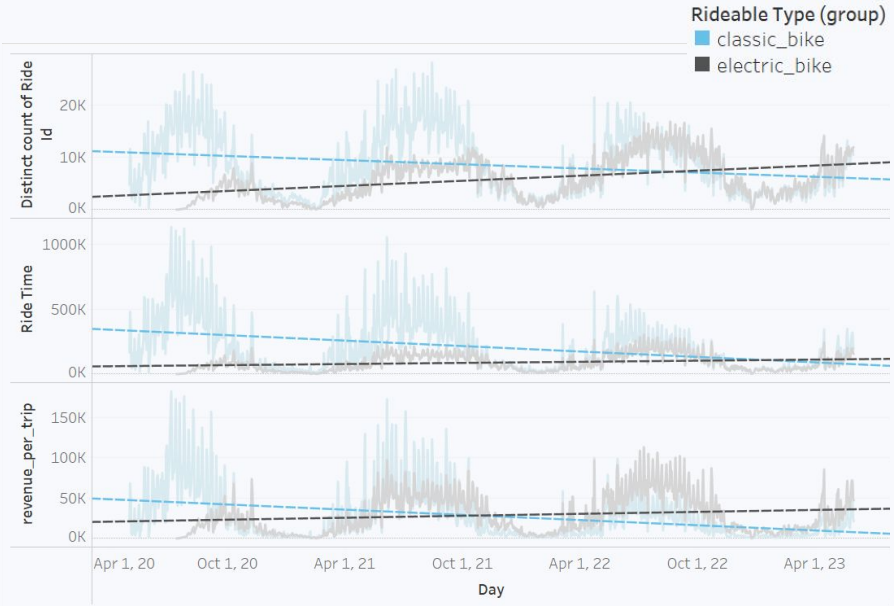
Based on the comparison of our data, we may **conclude that the season in Chicago** is the factor that causes the time series' fluctuation or trend.



Based on pareto, we can focus on the months that have the bigest potential income to the company there are from **may – october**

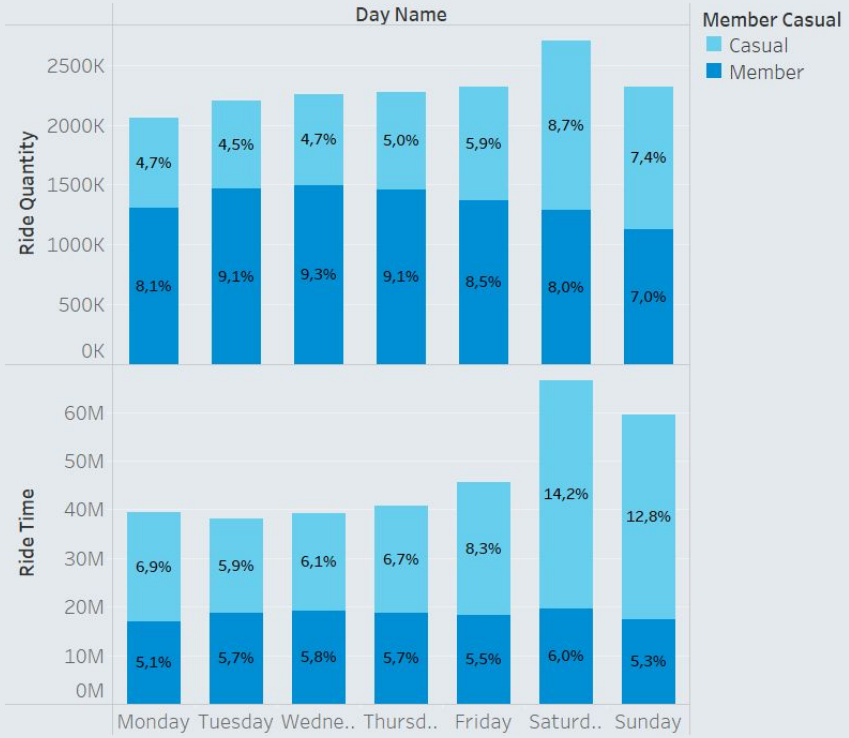
Bikes Trend by parameter

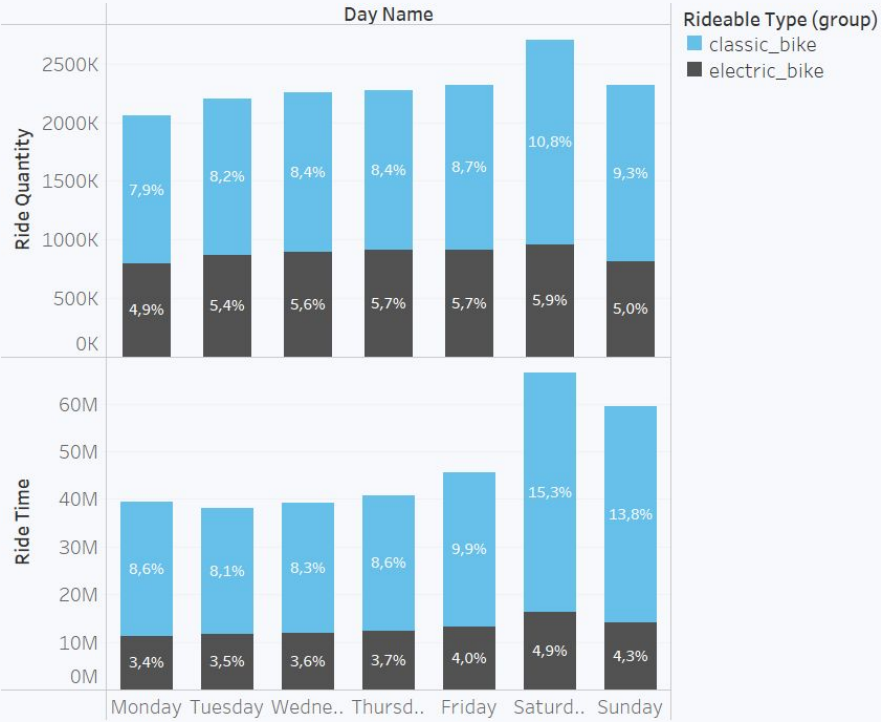




- Both of the user type ride quantity is increasing through time
- The ride time for casual user continues to drop while the ride time of the member is constant
- As the result, the revenue from casual member is continues to drop, while the revenue for member user is slightly increased

- Users are beginning to choose electric bikes over classic bikes as the trend for using classic bikes continues to drop while the trend for using electric bikes is rising.

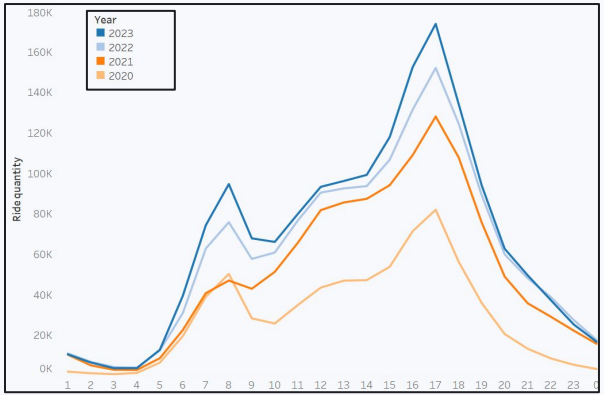




- According to the analysis done on a daily basis, we can see that renting and using shared bikes i**ncreases on Saturdays and Sundays.**
- The increase occurred in users with the **casual user category**
- In **member users**, the highest number of uses is actually on **weekdays**

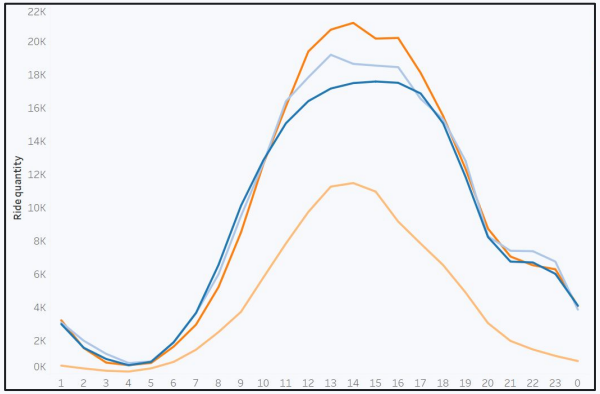
- There is no change in the number of people who ride electric bikes; nevertheless, the number of people who ride **classic bikes has increased**.

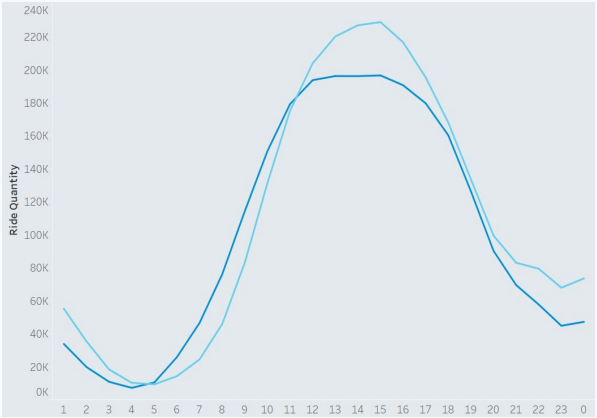
Weekday



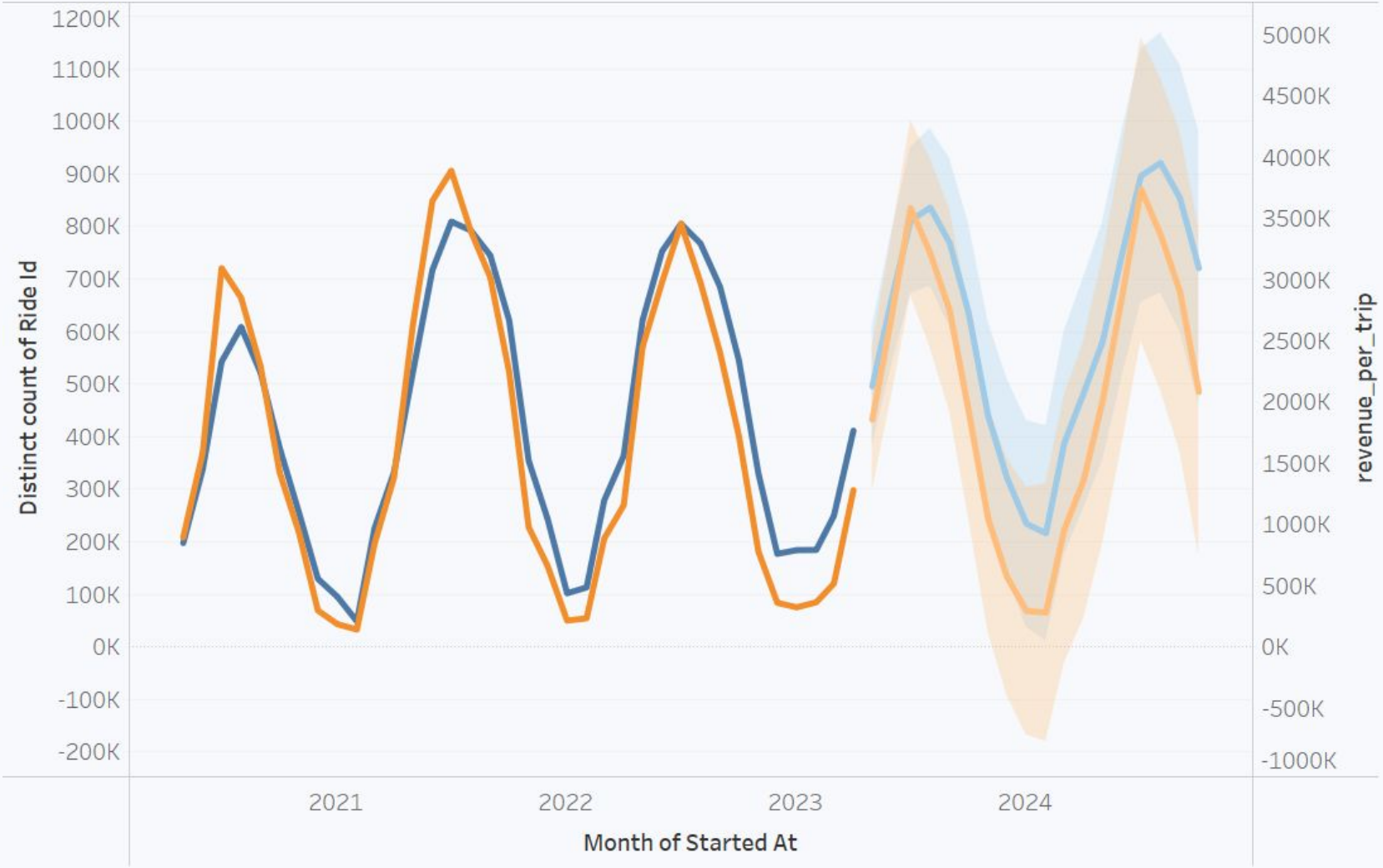


Weekend





- The lowest amount of bicycle usage was from 2am to 4am.
- There are differences in bicycle rental patterns by hour between weekends and weekdays.
- On **weekdays** the line graph fluctuates where the initial increase in bicycle rentals is at 8am and drops slightly at **9 & 10 am** then rises again in the next hour until it **peaks at 5pm** and then drops in the next hour.
- For weekends the graph is less volatile, rentals continue to increase starting from 6am and continue to increase slowly until 5pm and then decrease slowly until 11pm.
- On weekends we see that casual users **start renting bicycles from 7am and continue to grow until the number exceeds member users at 11am.**


1200K
1100K
1000K
900K
800K
700K
600K
500K
400K
300K
200K
100K
0K
-100K
-200K
Distinct count of Ride Id
5000K
4500K
4000K
3500K
3000K
2500K
2000K
1500K
1000K
500K
0K
-500K
-1000K
revenue_per_trip
2021
2022
2023
2024
Month of Started At

BUSSINESS QUESTIONS	Answers
• How often riders use the shared bikes within a given time frame?	There are **16,135,915** bicycle uses in the period 2020 to May 2023 which is divided into 22% in 2020, **34% in 2021, 34% in 2022** and 10% in 2023.
• Which groups are the most frequent users of our shared bikes, and how are the characteristics of each group?	**Member users** are the users with the highest amount of **bicycle usage**, but **casual users** are the users with the highest average usage time and **generate more revenue per trip** than member users.
• Which type of bikes that are the most frequent users of our shared bikes, and how are the characteristics of each group?	**Classic bikes** have the **highest usage** compared to electric bikes, but **electric bikes** are also **increasing in usage over time** and generate more revenue than classic bikes starting in 2022.
• Are there specific areas or stations where our shared bikes are in high demand? How can we optimize bike distribution to meet this demand?	**Streeter & Grand Ave, university ave, kingsbury st and wells st are popular station**s. To optimize this, we can increase the quota of bicycle slots at these locations by removing bicycles from the low usage location
• What are the most popular trip durations, and how can we cater to the needs of riders based on their preferred trip lengths?	**The most common duration is short duration < 15 minutes** and seems to be used only for commuting and not for recreation. We can facilitate this by ensuring docking stations are strategically located and functioning properly.
• Are there certain days or times when our shared bikes experience higher usage?	The increase in shared bikes usage is lowest in the winter period and peaks in the summer period.

Conclusions

- **The seasonality trend of user ride quantity and ride time is influenced by the season in chicago**, where it has the lowest number at winter and has it's peak at summer
- The revenue is **not solely depend on ride quantity but also by ride time**. thus, because of the median/average time is decreasing by time so is the revenue

Recommendations

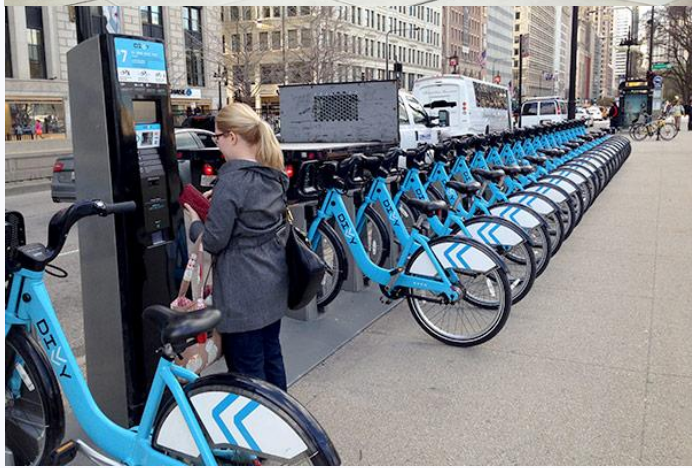
- Allocate resources to further **promote bike sharing** in order to maintain or increase the upward trend in bicycle usage: As an example is create a biking eventwith collaborations bike community or creating a healthy blog or content about health to body and environment

- **Focus on expanding bike sharing services in regions with the highest demand,** such as Streeter & Grand Ave, University Ave, Kingsbury St, and Wells St with

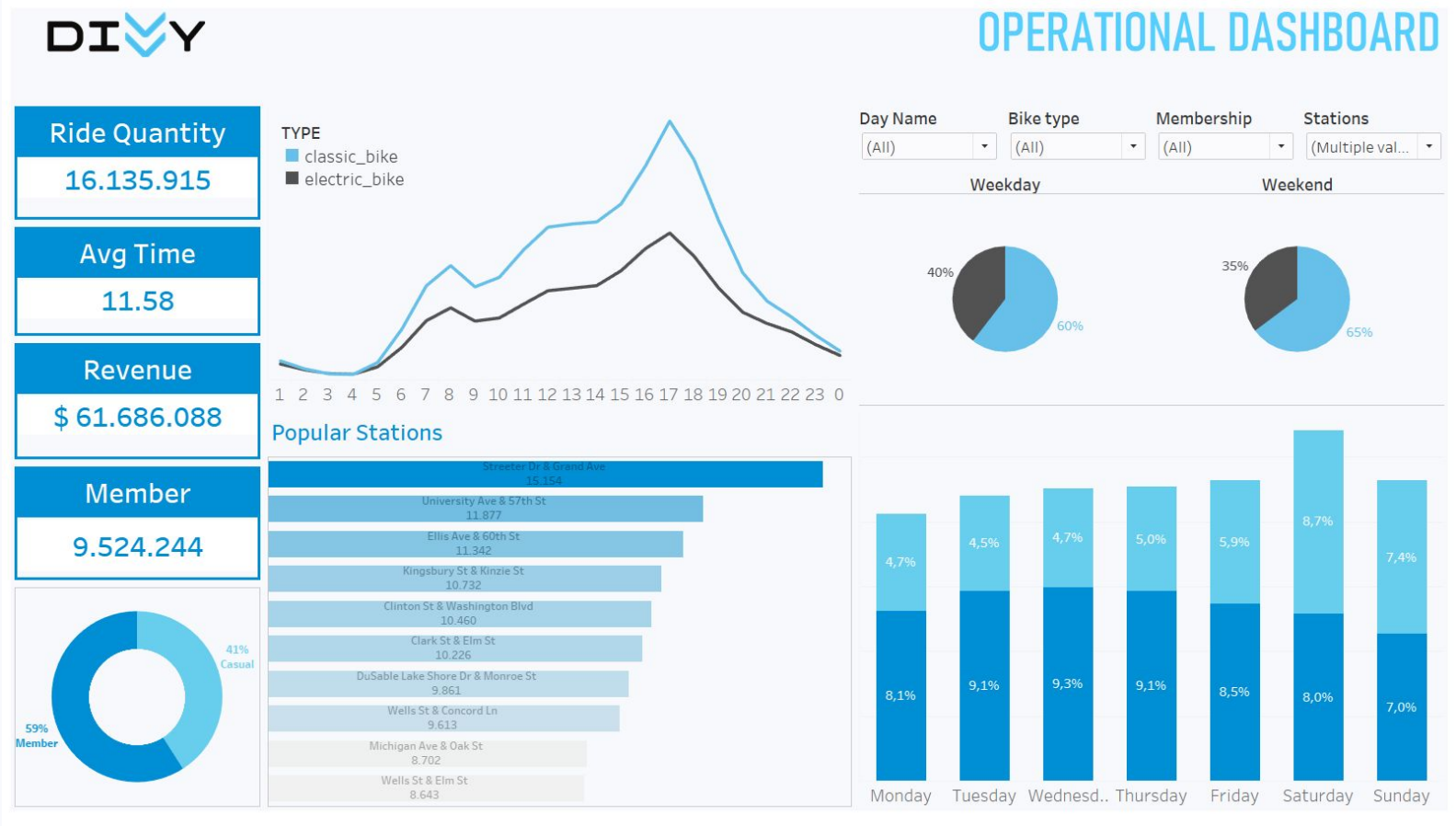
- Develop marketing campaigns and incentives to **maximize** the potential revenue **in summer period**, when bike usage is at the highest

- Monitor the usage patterns of electric bikes and consider increasing their availability (**Replace the missing/broken bikes with electric bikes**), as they generate more revenue than classic bikes starting in 2022





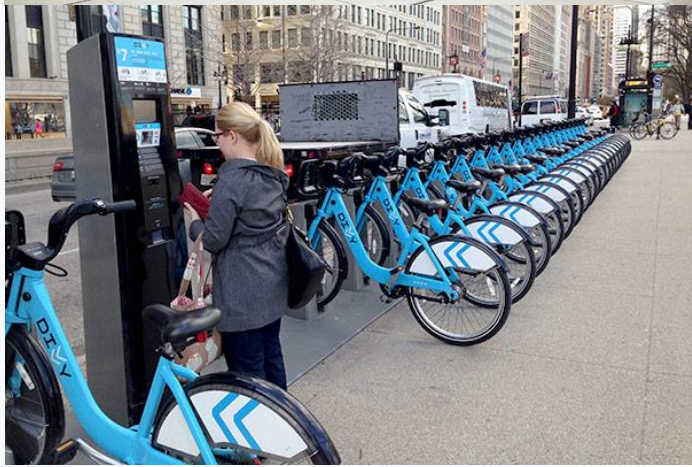
Dashboard


DIVVY
OPERATIONAL DASHBOARD
Ride Quantity
16.135.915
Avg Time
11.58
Revenue
$ 61.686.088
Member
9.524.244
TYPE
classic_bike
electric_bike
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 0
Day Name
(All)
Bike type
(All)
Membership
(All)
Stations
(Multiple val...
Weekday
Weekend
40%
60%
35%
65%
41%
Casual
59%
Member
Popular Stations
Streeter Dr & Grand Ave
15.154
University Ave & 57th St
11.877
Ellis Ave & 60th St
11.342
Kingsbury St & Kinzie St
10.732
Clinton St & Washington Blvd
10.460
Clark St & Elm St
10.226
DuSable Lake Shore Dr & Monroe St
9.861
Wells St & Concord Ln
9.613
Michigan Ave & Oak St
8.702
Wells St & Elm St
8.643
4,7%
8,1%
4,5%
9,1%
4,7%
9,3%
5,0%
9,1%
5,9%
8,5%
8,7%
8,0%
7,4%
7,0%
Monday
Tuesday
Wednesd..
Thursday
Friday
Saturday
Sunday

Idea for next analysis

- Divvy Station Dock Capacity Time Series Forecast
- Network Analysis
- Cluster Analysis





Thanks!

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Appendices